

Do **not** write solutions on this page.

12. [Maximum mark: 20]

- (a) Let $f(x) = (1 - ax)^{-\frac{1}{2}}$, where $ax < 1$, $a \neq 0$.

The n^{th} derivative of $f(x)$ is denoted by $f^{(n)}(x)$, $n \in \mathbb{Z}^+$.

Prove by induction that $f^{(n)}(x) = \frac{a^n (2n-1)! (1-ax)^{-\frac{2n+1}{2}}}{2^{2n-1} (n-1)!}$, $n \in \mathbb{Z}^+$. [8]

- (b) By using part (a) or otherwise, show that the Maclaurin series for $f(x) = (1 - ax)^{-\frac{1}{2}}$ up to and including the x^2 term is $1 + \frac{1}{2}ax + \frac{3}{8}a^2x^2$. [2]

- (c) Hence, show that $(1 - 2x)^{-\frac{1}{2}}(1 - 4x)^{-\frac{1}{2}} \approx \frac{2 + 6x + 19x^2}{2}$. [4]

- (d) Given that the series expansion for $(1 - ax)^{-\frac{1}{2}}$ is convergent for $|ax| < 1$, state the restriction which must be placed on x for the approximation $(1 - 2x)^{-\frac{1}{2}}(1 - 4x)^{-\frac{1}{2}} \approx \frac{2 + 6x + 19x^2}{2}$ to be valid. [1]

- (e) Use $x = \frac{1}{10}$ to determine an approximate value for $\sqrt{3}$.
Give your answer in the form $\frac{c}{d}$, where $c, d \in \mathbb{Z}^+$. [5]

